

Covariance, Correlation Coefficient and Index Numbers

A discussion

Noara Razzak

May 9, 2020

Covariance

It is a measure of how much two random variables vary together. It is similar to variance, but where variance tells you how a single variable varies, covariance tells you how two variables vary together. Covariance is denoted by $Cov(x, y)$.

Correlation Coefficient

It is used to test relationships between quantitative variables or categorical variables. In other words, it's a measure of how things are related. The study of how variables are correlated is called correlation analysis. Correlation coefficient is denoted by r .

Covariance and Correlation Coefficient

Suppose the sales of ice cream per day is dependent on temperature. The data is given in the table.

Covariance and Correlation Coefficient

Temp (C)	Sales(10^3 units)	$x - \bar{x}$	$y - \bar{y}$	$x - \bar{x} \cdot y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
21	10	-7.5	-8.33	62.48	56.25	69.39
24	11	-4.5	-7.33	32.985	20.25	53.73
27	13	-1.5	-5.33	7.995	2.25	28.41
30	19	1.5	0.67	1.005	2.25	0.45
33	25	4.5	6.67	30.02	20.25	44.49
36	32	7.5	13.67	102.53	56.25	186.87

Covariance and Correlation Coefficient

Finding Covariance

If we sum the 5th column we get the covariance of x and y , which is $Cov(x, y) = \sum (x - \bar{x}) * (y - \bar{y}) = 237$. Hence, the covariance in this problem is 237.

Covariance and Correlation Coefficient

Finding Variances

If we sum the 6th column we get the variance of x , which is given by

$$\text{Var}(x) = \sum (x - \bar{x})^2 = 157.5$$

If we sum the 7th column we get the variance of y , which is given by

$$\text{Var}(y) = \sum (y - \bar{y})^2 = 383.3$$

Covariance and Correlation Coefficient

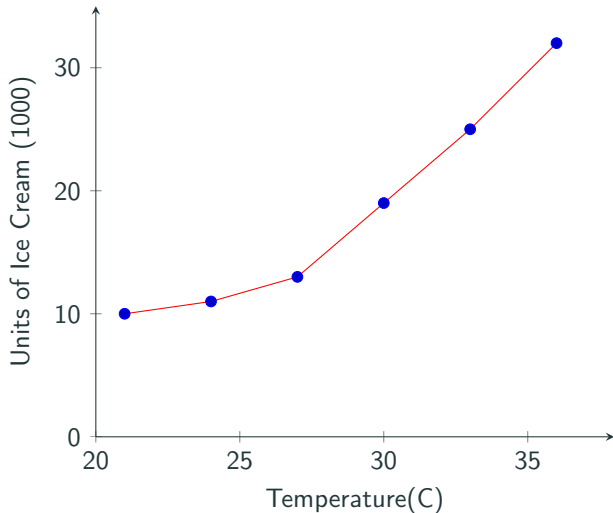
Finding Correlation Coefficient

If we divide the covariance of x and y by the standard deviation of each x and y we will get:

$$\begin{aligned} r &= \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x) \cdot \text{Var}(y)}} \\ &= \frac{(\sum x - \hat{x}) * (y - \hat{y})}{\sqrt{\sum (x - \hat{x})^2 \cdot \sum (y - \hat{y})^2}} \\ &= \frac{237}{\sqrt{157.7} \cdot \sqrt{383.3}} \\ &= 0.9645 \end{aligned}$$

Covariance and Correlation Coefficient

An illustration



Break

*We have finished discussing covariance and correlation coefficient,
now we will move onto index numbers*

Index Numbers

An index starts with a base value, typically set at 100, regardless of whether the index measures data units in dollars, euros, or headcount, for example. Each subsequent value in the index is then normalized to this base value.

Index Numbers

Calculating Index Numbers

The first step in constructing an index involves setting the base value. For a time series of annual price of a bundle of commodities, for example, say the first year, prices were Taka 150. This base-year amount is set to equate to the starting index value of 100. Each added value becomes normalized against the base value.

Index Numbers

Calculating Index Numbers

To calculate the value of the next data point in this indexed time series, let us say the second year of commodity prices equates to Taka 225. You would divide the new data point (Taka 225) by the original one (Taka 150), multiplying the result by 100 as follows to get a year 2 index value of 167.

$$\frac{(\text{Year 2 price : 225})}{(\text{Year 1 price : 150})} \cdot 100 = 167$$

Index Numbers

Example: An index of house prices in Banani, Dhaka

Year	Average House Prices (Lakhs)	Percentage Change	Index
2013	200	Base Year	100
2014	220	10	110
2015	230	15	115
2016	210	5	105
2017	195	-2.43	97.57

Index Numbers

This is the end of our syllabus. Good luck with the final quiz!